

on the resolution of the printer. Just as high resolution paper printers give us a clearer and crisper final image, higher resolution 3D printers are able to craft objects of great precision and accuracy. A 3D printer's resolution is calculated on the horizontal two-dimensional X-Y axis in dots per inch (DPI). The thickness of each layer is generally 250 DPI which is about 0.1 millimeter (100 μm), but can also be as little as 16 μm or 0.00016 millimeter. The individual particle or dot size produced by these printers is on average between 50 μm to 100 μm in diameter.

Additive printing methodologies rely heavily on the accuracy and detailing of the particulate size chosen to achieve perfect fabrication. The time and effort taken by a 3D printer is therefore based entirely on the scale, size and complexity of the object being created and the specific printer being used. Some printers are about the size of a microwave oven while others can take up a whole room.

The finishing process

The finishing process, as we discussed in the steps, is a careful and delicate process, but usually the fastest in the entire workflow. What's more, some basic finishing techniques allow for greater quality in the final product. Though 3D printers that can print in high resolution are available, it's possible to achieve even higher quality through a post-printing subtractive process.

While additive printing is about adding materials together, subtractive printing is the exact opposite. This process is similar to photographers shooting an image at the highest resolution, then cropping out the desired part and rescaling it to fit a smaller frame. This allows for greater clarity and quality in the image. In 3D printing, a slightly larger or oversized version of the item is printed



Object finishing is critical if you want a smooth surface

at normal resolution and then subjected to a high-resolution subtractive process, which makes it possible to craft a more accurately sized final object. This is akin to scaling down a larger sized model proportionally to achieve greater precision in detail.